

ODCR: Reliability-Gated Dynamic Causal Refinement for Cross-Domain Explainable Recommendation

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Abstract

Cross-domain explainable recommendation must predict target-domain preferences and generate explanations that remain content-specific, domain-appropriate, and faithful to recommendation evidence. A central failure mode is explanation degeneration, where generators learn self-reinforcing textual shortcuts instead of user–item rationale. Building on D4C, which diagnoses degeneration as a dynamic feedback process $W^{(t)} \rightarrow E^{(t)} \rightarrow W^{(t+1)}$, ODCR refines the coarse textual attribute $W^{(t)}$ into stable content evidence C , feedback-sensitive hierarchical style $S^{(t)}$, and counterfactual reliability R_{cf} . This refinement yields a complete control loop: evidence-anchored content/style decomposition, hierarchical style state modeling, reliability-gated counterfactual routing, and controlled faithful explanation generation. ODCR therefore learns explanations from $P(E \mid U, I, C, S, R_{cf})$ rather than shortcut dependence on $P(E \mid W)$. We report current single-run results on four directed transfer settings, treating them as bounded evidence for the proposed dynamic causal refinement rather than as a multi-seed benchmark-dominance claim.

1 Introduction

Explainable recommendation asks a system to predict user preferences and to justify those predictions with natural-language explanations [7]. In cross-domain settings this problem becomes especially fragile: evidence can transfer across domains, but review conventions, item vocabularies, and explanation styles can shift at the same time [8]. A generator that appears fluent in the target domain may still produce explanations that are generic, repetitive, or weakly connected to the user–item evidence. We refer to this failure mode as explanation degeneration.

D4C provides the closest causal foundation for this problem [6]. Its key insight is not only that hidden textual attributes W can act as domain-level shortcuts, but that these shortcuts participate in a dynamic feedback process. Textual attributes at time t influence explanations, and generated or observed explanations can reshape future textual attributes:

$$W^{(t)} \rightarrow E^{(t)} \rightarrow W^{(t+1)}.$$

This dynamic view explains why degeneration can be self-reinforcing. Ordinary generators can learn a shortcut from $W^{(t)}$ to $E^{(t)}$, preserving domain-looking language while weakening the genuine user–item interaction path that should support an explanation.

ODCR builds on this dynamic feedback diagnosis. Its starting point is that a single coarse textual variable $W^{(t)}$ is too broad for controlled explanation learning. The variable mixes at least three forces: stable content evidence, feedback-sensitive expression style, and the reliability of counterfactual evidence. ODCR therefore refines the D4C textual-feedback view as

$$W^{(t)} \Rightarrow (C, S^{(t)}, R_{cf}),$$

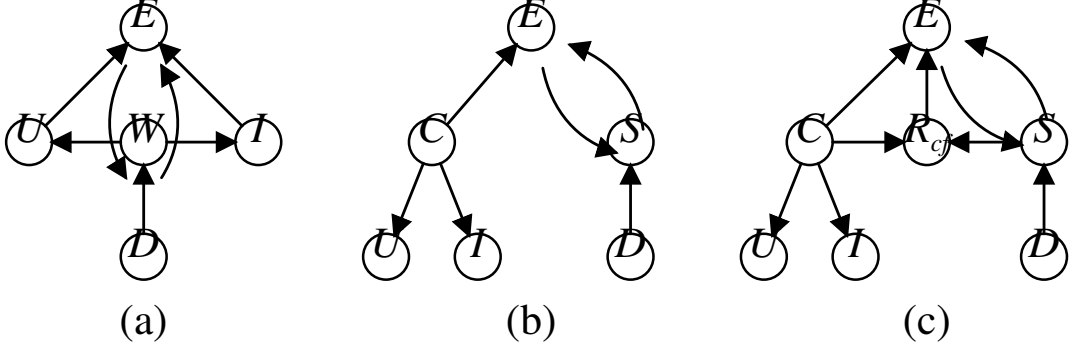


Figure 1: Dynamic causal refinement from D4C to ODCR. D4C explains explanation degeneration as a feedback loop between contextual textual attributes and explanations. ODCR refines this loop into stable content evidence, feedback-sensitive hierarchical style, and counterfactual reliability control.

where C denotes stable content evidence, $S^{(t)}$ denotes a dynamic hierarchical style state, and R_{cf} denotes the reliability with which counterfactual evidence should influence learning. The resulting method is not an engineering-stage pipeline. It is a reliability-gated dynamic causal control framework for cross-domain explainable recommendation.

Figure 1 summarizes the refinement. ODCR first anchors content/style decomposition in evidence, so that content C captures what should be explained while style $S^{(t)}$ captures how the explanation is expressed. It then models style hierarchically as a domain-global state plus instance-local residual, localizing dynamic feedback to style rather than allowing feedback to overwrite content. Counterfactual examples are further filtered and weighted by R_{cf} , so unreliable shifts do not dominate learning. Finally, the explanation generator is conditioned on (U, I, C, S, R_{cf}) , with faithfulness alignment keeping generated content consistent with rating-relevant evidence.

This manuscript makes four contributions:

- **Dynamic causal refinement.** ODCR builds on D4C’s dynamic textual-feedback diagnosis and refines coarse textual attributes into controllable variables for content, style, and counterfactual reliability.
- **Evidence-anchored decomposition.** ODCR separates stable content evidence from feedback-sensitive domain style, avoiding an undifferentiated textual shortcut.
- **Reliability-gated counterfactual control.** ODCR filters, routes, and weights counterfactual evidence using reliability and uncertainty-aware signals.
- **Controlled faithful generation.** ODCR generates explanations conditioned on content, style, and reliability-controlled evidence while keeping explanation content aligned with rating-relevant evidence.

2 Related Work

2.1 Generative Explainable Recommendation and Degeneration

Explainable recommendation studies models that pair recommendation decisions with textual or structured rationales [7]. Neural and language-model-based systems such as PETER, PEPLER,

ReXPlug, ERRa, and AdaReX demonstrate that explanations can be generated jointly with recommendation signals [3, 4, 2, 1, 5]. These models also highlight the central risk for this paper: fluent explanations are not automatically faithful explanations. When a generator overuses common review phrases or domain templates, explanation quality degenerates even if the text looks plausible.

2.2 Cross-Domain Explainable Recommendation

Cross-domain recommendation transfers evidence from a source domain to a target domain [8]. Cross-domain explainable recommendation adds a second transfer problem: the explanation must preserve recommendation content while adapting to target-domain language. This makes content and style inseparable if the model treats text as one coarse signal. ODCR addresses this by representing content evidence and style state as separate control variables rather than relying on a single domain-level textual attribute.

2.3 Causal and Counterfactual Recommendation

D4C is the closest causal foundation for ODCR [6]. It diagnoses cross-domain explanation degeneration through hidden textual attributes W and uses domain counterfactual reasoning to reduce shortcut dependence. ODCR inherits this causal view but emphasizes the dynamic feedback part of the diagnosis: $W^{(t)}$ influences explanations, and explanations can feed back into future textual attributes. ODCR refines this process into stable content evidence C , dynamic style $S^{(t)}$, and reliability-gated counterfactual influence R_{cf} .

2.4 Controlled Generation, Faithfulness, and Consistency

Controlled explanation generation requires more than producing text in the right domain style. The generated explanation should remain tied to the content evidence that supports the recommendation. ODCR therefore positions controlled generation and faithfulness alignment as output-side parts of the method: C controls what is said, $S^{(t)}$ controls how it is expressed, and R_{cf} controls which counterfactual signals are trusted during learning. Modern external systems such as CIER, MAPLE, ELIXIR, Prism, and XRec remain comparison targets once adapted to the same data and evaluation setting; the current manuscript does not claim results against them.

3 Method

3.1 Problem Formulation

Let $\mathcal{D}_s$ and $\mathcal{D}_t$ denote a source domain and a target domain. Each domain contains user-item interaction records (u, i, r, e) , where u is a user, i is an item, r is an observed rating, and e is a natural-language explanation or review-derived explanation target. Given source-domain evidence and target-domain evidence, the goal is to predict a target-domain rating $\hat{r}_{ui}$ and generate a target-domain explanation $\hat{e}_{ui}$ for a user-item pair.

The target explanation should satisfy two constraints. First, it should be content-grounded: the explanation should describe evidence relevant to the user and item rather than generic domain text. Second, it should be domain-appropriate: the explanation should use target-domain expression style without allowing style to replace recommendation evidence. ODCR formalizes these constraints through content evidence C , dynamic style $S^{(t)}$, and counterfactual reliability R_{cf} .

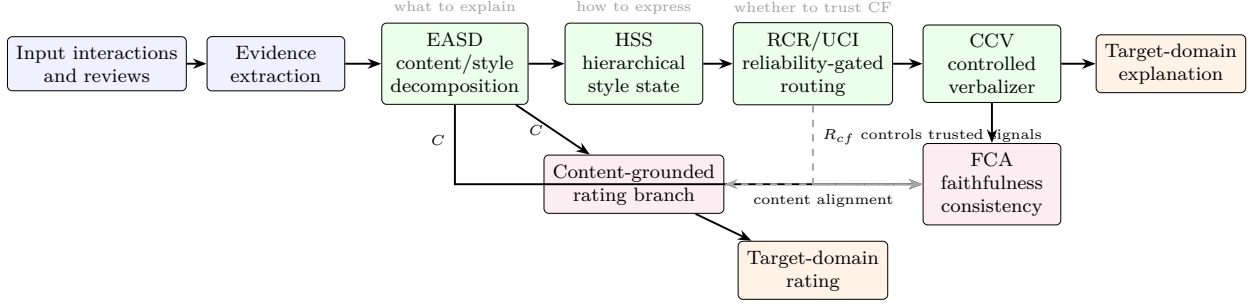


Figure 2: ODCR overall architecture. The method maps input interactions and reviews into evidence extraction, EASD content/style decomposition, HSS hierarchical style state modeling, RCR/UCI reliability-gated counterfactual routing, CCV controlled explanation generation, and FCA faithfulness consistency. The content-grounded rating branch is separated from explanation generation while sharing content evidence.

3.2 Dynamic Feedback View of Explanation Degeneration

D4C diagnoses explanation degeneration as a causal shortcut involving hidden textual attributes W [6]. In the static reading, W captures domain-level textual factors that can affect users, items, and explanations. The higher-value reading for ODCR is dynamic: textual attributes and generated explanations can reinforce one another over time,

$$W^{(t)} \rightarrow E^{(t)} \rightarrow W^{(t+1)}.$$

When this loop is left uncontrolled, a generator can learn the shortcut $W^{(t)} \rightarrow E^{(t)}$ instead of the intended interaction path $(U, I) \rightarrow E^{(t)}$. The resulting explanation may fit the target domain’s surface language while failing to explain the specific user–item preference.

This feedback view motivates a control problem. The model should preserve the part of textual evidence that supports the recommendation, allow style to adapt across domains, and prevent unreliable counterfactual shifts from being treated as trustworthy training evidence. ODCR keeps D4C’s causal diagnosis but replaces the coarse textual shortcut with explicit control variables.

3.3 From Coarse Textual Attributes to ODCR Control Variables

ODCR refines the D4C textual attribute at time t as

$$W^{(t)} \Rightarrow (C, S^{(t)}, R_{cf}).$$

Here C is stable content evidence: it captures what should be explained and what can support rating prediction. $S^{(t)}$ is a dynamic style state: it captures how content is expressed in a domain at time t . R_{cf} is counterfactual reliability: it controls whether and how strongly counterfactual evidence should influence learning.

Figure 2 shows the resulting architecture. ODCR does not ask one representation to absorb every textual factor. Instead, it uses C , $S^{(t)}$, and R_{cf} as method-level controls for content, expression, and counterfactual trust.

3.4 Evidence-Anchored Content/Style Decomposition

Evidence-Anchored Content/Style Decomposition (EASD) is the first ODCR module. It separates stable content evidence from expression style while keeping both anchored to observable user–item

evidence. Content evidence C is tied to rating-relevant aspects and interaction evidence. Style evidence captures domain phrasing, emphasis, and lexical conventions that affect how content is verbalized.

The key distinction is that ODCR does not leave shared and specific representations to infer this split without guidance. The decomposition is anchored: content is constrained by preference evidence, while style is constrained by domain-specific expression signals. This turns the coarse attribute $W^{(t)}$ into a controllable pair: C determines what should be explained, and $S^{(t)}$ determines how that evidence is expressed. Rating prediction and explanation generation can therefore share content evidence without collapsing explanation style into the rating branch.

3.5 Hierarchical Style State

Hierarchical Style State (HSS) models style as more than a domain identifier. ODCR writes the style state as

$$S^{(t)} = (S_{\text{domain}}^{(t)}, \Delta S_{\text{local}}^{(t)}),$$

where $S_{\text{domain}}^{(t)}$ captures domain-global textual conventions and $\Delta S_{\text{local}}^{(t)}$ captures instance-local residual style. The domain component models broad conventions such as review tone and domain-common phrasing. The local residual captures user-item-specific stylistic variation that should not be forced into a single domain template.

HSS makes the D4C feedback channel more precise. If explanations feed back into the future textual environment, ODCR localizes that feedback mainly to $S^{(t+1)}$. Content evidence C should remain a stable anchor for what is being recommended and explained. This prevents dynamic feedback from corrupting content while still allowing expression style to evolve with the target domain.

3.6 Reliability-Gated Counterfactual Control

Reliability-Gated Counterfactual Control combines RCR and uncertainty-aware counterfactual invariance (UCI). D4C shows that counterfactual domain evidence can reduce shortcut dependence, but ODCR further recognizes that counterfactual examples are not equally trustworthy. Some preserve content and shift only style; others alter the recommendation evidence or introduce noisy signals.

ODCR uses two control layers. The first layer is eligibility filtering: a counterfactual example is usable only if it preserves the content needed for the target explanation and does not violate the intended transfer setting. The second layer is reliability weighting and sampling priority: usable counterfactual examples receive different influence according to R_{cf} and uncertainty. A compact module-level objective is

$$\mathcal{L}_{\text{ODCR}} = \mathcal{L}_{\text{fact}} + \lambda \sum_j R_{cf,j} \mathcal{L}_{cf,j},$$

where $\mathcal{L}_{\text{fact}}$ is factual learning, $\mathcal{L}_{cf,j}$ is the counterfactual learning signal for candidate j , and $R_{cf,j}$ controls its influence. UCI reduces trust in unstable or uncertain counterfactual signals rather than treating every counterfactual example as equally informative.

Figure 3 illustrates this two-layer routing mechanism. The figure is intended as a method diagram and as an insertion point for later module ablations; it does not claim that a complete ablation suite is already integrated into the current result tables.

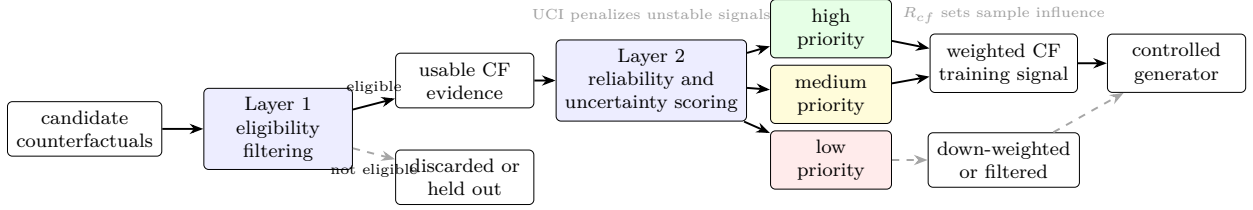


Figure 3: RCR/UCI routing mechanism. Counterfactual candidates first pass eligibility filtering, then receive reliability and uncertainty-aware weights that determine sampling priority and training influence. Low-reliability evidence is down-weighted or filtered rather than treated as equal counterfactual supervision.

3.7 Controlled Causal Verbalizer and Faithfulness Alignment

The Controlled Causal Verbalizer (CCV) is the output-side module. Instead of learning the shortcut $P(E | W)$, ODCR conditions explanation generation on the refined variables:

$$P(E | U, I, C, S, R_{cf}).$$

Content evidence C controls what should be said, style state S controls how it should be expressed, and R_{cf} controls which counterfactual training signals are trusted. This design makes explanation generation a controlled causal verbalization problem rather than an unconstrained domain-language imitation problem.

Faithfulness Consistency Alignment (FCA) keeps the generated explanation tied to content-grounded recommendation evidence. Style transfer should not substitute for preference evidence, and a fluent target-domain explanation should not drift away from the aspects that support the rating branch. FCA is therefore the consistency principle linking CCV back to C : the output may adapt its expression style, but its content should remain aligned with rating-relevant evidence.

3.8 Implementation Mapping

The implementation maps these method modules to the current ODCR workflow without making workflow stages the paper’s primary contribution. Preprocessing and Step3 support EASD/HSS by preparing evidence and learning content/style representations. Step4 supports RCR/UCI through reliability estimation, routing, confidence buckets, uncertainty signals, and sample-weight hints. Step5_e supports CCV/FCA as explanation-only controlled generation. Rating evidence remains content-grounded and separate from explanation generation, while explanation learning consumes reliability-gated evidence for target-domain verbalization.

4 Experimental Setup

4.1 Datasets and Directed Transfer Settings

ODCR is evaluated on four directed cross-domain explainable recommendation settings. The strong-related Amazon settings transfer between AM-Movies and AM-CDs in both directions. The weak cross-platform settings transfer between TripAdvisor and Yelp in both directions. In each setting, the dataset named in the result-table block is the evaluation dataset, and the auxiliary domain is the domain used for cross-domain transfer.

4.2 Metrics

We report automatic text-generation metrics and rating-error metrics in the same overall performance tables. The text metrics are ROUGE-1, ROUGE-L, BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, DIST-1, and DIST-2. Rating errors are reported by RMSE and MAE. Larger values are better for the automatic text metrics, while smaller values are better for RMSE and MAE.

4.3 Baselines and Comparison Scope

The comparison includes single-domain methods (DualPC, PETER, ReXPlug, and ERRa) and cross-domain methods (PEFT, PEPLER, AdaReX, D4C, and ODCR). D4C is the closest causal reference point for ODCR’s dynamic textual-feedback view [6]. Classical explainable recommendation systems are cited as task-family context where appropriate [3, 4, 5]. Modern external baselines such as CIER, MAPLE, ELIXIR, Prism, and XRec remain future comparison targets unless they are adapted to the same data and evaluation setting.

4.4 Evaluation Scope

We report single-run results on four directed transfer settings. Baseline values are adapted from the original D4C reporting and are shown without standard deviations for compactness. ODCR values are current single-run results, so the tables should be read as bounded evidence rather than as a multi-seed benchmark claim.

5 Results

5.1 Overall Performance on Strong-Related Amazon Tasks

The strong-related Amazon setting evaluates transfer between two related product domains. The AM-CDs block uses AM-Movies as the auxiliary domain, and the AM-Movies block uses AM-CDs as the auxiliary domain. This setting is the natural case for ODCR’s content/style separation: related domains can share some content evidence while still requiring domain-specific expression style. The table therefore should be read jointly across rating and explanation metrics. RMSE and MAE summarize rating behavior, while ROUGE, BLEU, METEOR, and diversity metrics summarize automatic explanation behavior. ODCR’s current single-run values show stronger automatic text metrics in this setting while retaining explicit rating-error reporting in the same table.

5.2 Overall Performance on Weak Cross-Platform Tasks

The TripAdvisor–Yelp setting is a weaker cross-platform transfer case. Here the style shift is larger, and counterfactual evidence is more likely to vary in reliability. This makes the setting a stress test for the ODCR control loop: EASD must preserve content evidence, HSS must represent a larger style shift, RCR/UCI must avoid over-trusting unstable counterfactual signals, and CCV/FCA must generate explanations that remain content-grounded. The table again reports rating errors and automatic explanation metrics together rather than reducing the comparison to a single score.

5.3 Discussion of the Results

Across the two tables, the main result presentation is organized by dataset block and method comparison rather than by implementation stages. This aligns the evidence with the paper’s dynamic causal story. The Amazon blocks are best interpreted as related-domain content/style transfer,

Table 1: Overall Performance on the Strong-Related Amazon Tasks

Methods	ROUGE-1	ROUGE-L	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	DIST-1	DIST-2	RMSE	MAE
<i>AM-CDs Dataset, auxiliary domain: AM-Movies</i>											
Single Domain											
DualPC	12.50	9.67	9.32	3.69	1.60	0.88	7.98	0.01	0.02	0.9563	0.6736
PETER	11.62	9.07	10.42	3.86	1.77	0.99	8.41	0.78	2.64	0.9191	0.6577
ReXPlug	13.16	10.18	9.85	3.86	1.64	0.88	8.13	0.59	2.53	0.9580	0.6751
ERRA	12.72	9.92	11.53	4.32	1.99	1.12	8.88	0.42	1.46	0.9128	0.6592
Cross-Domain											
PEFT	12.41	9.69	11.67	4.29	1.95	1.10	8.98	0.53	1.75	0.9050	0.6767
PEPLER	12.80	9.71	11.67	4.07	1.78	0.97	8.94	0.70	2.94	0.9003	0.6568
AdaReX	13.38	10.40	12.60	4.67	2.09	1.15	9.52	0.89	3.13	0.9178	0.6582
D4C	13.41	10.49	12.84	4.85	2.29	1.26	9.82	0.98	4.07	0.9179	0.6601
ODCR	22.6185	15.4686	17.1937	8.6939	4.9249	3.0127	16.5699	1.8254	11.1427	0.8494	0.5764
<i>AM-Movies Dataset, auxiliary domain: AM-CDs</i>											
Single Domain											
DualPC	11.83	9.10	9.21	3.04	1.23	0.63	7.94	0.03	0.06	1.0504	0.7567
PETER	14.35	10.92	13.24	4.82	2.18	1.21	9.75	0.48	1.85	0.9959	0.7411
ReXPlug	10.86	8.30	6.83	2.10	0.72	0.30	6.55	0.39	1.37	1.0236	0.7479
ERRA	13.71	10.47	12.14	4.41	1.99	1.14	9.42	0.33	1.26	1.0025	0.7413
Cross-Domain											
PEFT	13.79	9.93	12.24	4.72	1.90	1.05	9.56	0.57	2.23	0.9944	0.7339
PEPLER	13.66	11.20	12.80	5.25	2.20	1.29	10.21	0.59	2.98	0.9919	0.7598
AdaReX	13.97	10.60	12.53	4.62	2.11	1.20	9.94	0.44	2.29	0.9987	0.7305
D4C	14.45	11.08	13.41	4.95	2.26	1.31	10.19	1.09	4.30	1.0045	0.7352
ODCR	26.0735	17.6785	20.1472	10.7957	6.3840	4.0431	19.3942	1.4477	9.5002	0.8715	0.5750

Note: For each dataset block, cross-domain methods use the auxiliary domain indicated in the block heading. Baseline values are reported without standard deviations. ODCR values are current single-run results.

while the TripAdvisor–Yelp blocks emphasize larger style mismatch and reliability control. The current ODCR rows provide bounded single-run evidence for the proposed method framing. They should not be read as state-of-the-art claims, statistical-significance claims, full D4C reproduction, or multi-seed benchmark dominance.

6 Discussion

6.1 Dynamic Feedback Control

ODCR treats explanation degeneration as a dynamic causal-control problem. D4C’s feedback view explains why domain textual shortcuts can become self-reinforcing: $W^{(t)}$ influences explanations, and explanations can feed back into $W^{(t+1)}$. ODCR does not discard this diagnosis. It refines the feedback loop so that stable content evidence, dynamic style, and counterfactual reliability are controlled separately.

6.2 Role of EASD and HSS

EASD and HSS are the parts of ODCR that keep the paper from becoming centered on reliability routing alone. EASD anchors content/style decomposition in evidence, separating what should be explained from how it is expressed. HSS makes style feedback hierarchical, with domain-global style and instance-local residuals. Together, they define where dynamic feedback should operate:

Table 2: Overall Performance on the Weak Cross-Platform Tasks

Methods	ROUGE-1	ROUGE-L	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	DIST-1	DIST-2	RMSE	MAE
<i>TripAdvisor Dataset, auxiliary domain: Yelp</i>											
Single Domain											
DualPC	16.41	12.59	14.80	5.62	2.10	0.85	10.35	0.04	0.09	0.8290	0.6260
PETER	16.72	12.79	15.64	5.85	2.43	1.09	10.46	0.58	1.89	0.8193	0.6205
ReXPlug	17.40	12.78	12.98	5.01	1.94	0.74	10.29	0.25	1.40	0.8571	0.6541
ERRA	16.76	13.14	16.07	5.92	2.47	1.10	10.10	0.61	2.24	0.7896	0.6091
Cross-Domain											
PEFT	15.93	12.16	14.21	5.34	2.25	1.00	9.20	0.22	0.81	0.8000	0.6180
PEPLER	16.07	11.72	13.26	4.82	2.00	0.87	9.59	0.28	1.54	0.7910	0.6084
AdaReX	16.77	13.18	16.27	6.20	2.48	1.13	10.31	0.60	1.82	0.7904	0.6058
D4C	17.08	13.18	16.70	6.22	2.57	1.15	10.76	0.71	2.59	0.8014	0.6132
ODCR	71.7019	52.5850	57.6942	40.3661	29.5623	22.3412	56.9855	3.7559	25.2687	0.7572	0.5748
<i>Yelp Dataset, auxiliary domain: TripAdvisor</i>											
Single Domain											
DualPC	10.48	8.11	10.33	2.71	0.95	0.41	6.31	0.05	0.04	1.0699	0.8264
PETER	12.34	9.55	12.30	3.76	1.47	0.62	7.87	0.79	2.65	1.0459	0.8095
ReXPlug	9.45	7.51	9.45	2.32	0.72	0.30	6.22	0.37	1.73	1.0414	0.8141
ERRA	11.56	9.02	10.66	3.24	1.29	0.58	7.00	0.47	2.49	1.0519	0.8106
Cross-Domain											
PEFT	11.32	8.79	11.54	3.39	1.30	0.56	7.37	1.20	3.90	1.0393	0.8085
PEPLER	12.13	9.19	10.67	3.38	1.39	0.60	8.07	0.51	3.84	1.0358	0.7984
AdaReX	12.36	9.61	12.31	3.78	1.49	0.64	7.98	0.60	3.05	1.0561	0.8050
D4C	13.22	10.29	12.75	4.03	1.62	0.72	8.23	1.22	3.95	1.0472	0.8075
ODCR	70.8476	53.2973	57.6096	40.8406	30.4034	23.3925	56.8649	2.8513	25.0243	0.9575	0.7301

Note: For each dataset block, cross-domain methods use the auxiliary domain indicated in the block heading. Baseline values are reported without standard deviations. ODCR values are current single-run results.

generated explanations may reshape style, but they should not overwrite content evidence.

6.3 Role of RCR/UCI and CCV/FCA

RCR/UCI controls counterfactual influence after decomposition. It filters counterfactual candidates, scores reliability, and uses uncertainty to reduce trust in unstable shifts. CCV/FCA then turns the controlled evidence into target-domain explanations. CCV conditions generation on U , I , C , S , and R_{cf} ; FCA keeps explanation content aligned with content-grounded rating evidence. This closes the ODCR loop from dynamic feedback diagnosis to controlled faithful generation.

6.4 Reading the Current Evidence

The current result tables provide a compact main-text comparison over four directed transfer settings. The Amazon blocks illustrate related-domain content/style transfer, while the TripAdvisor–Yelp blocks place more stress on style shift and reliability control. These results support the manuscript as bounded current evidence. Additional ablations, human assessment, multi-seed evaluation, and broader baseline adaptation can be integrated as later evidence without changing the core method story.

7 Limitations

Evaluation scope. The current manuscript reports current single-run ODCR values on four directed transfer settings. The tables should not be interpreted as seed-robust mean $\pm$ std estimates or as statistical-significance evidence.

Reference and benchmark compatibility. Strict longest-reference reconstruction and full D4C paper-compatible benchmark reproduction are outside the present paper boundary. The current tables are organized for compact method presentation, with baseline values adapted from D4C-style reporting and ODCR values shown as current bounded evidence.

Broader baselines. Modern external baselines such as CIER, MAPLE, ELIXIR, Prism, and XRec require same-setting adaptation before they can be used as result-table comparisons. They are therefore discussed as comparison scope rather than as current result claims.

Ablation integration. The method section exposes module-level hooks for EASD, HSS, RCR/UCI, and CCV/FCA. Ablation evidence can be connected to these hooks as it is finalized. Until then, the manuscript avoids assigning the observed table patterns to any single component in isolation.

Figure and presentation polish. The core method figures are now vector/TikZ diagrams intended for manuscript review. Their visual density, placement, and AAAI camera-ready styling may still need refinement after Chat review.

8 Conclusion

ODCR is presented as a reliability-gated dynamic causal refinement framework for cross-domain explainable recommendation. Starting from D4C’s feedback view of explanation degeneration, ODCR refines coarse textual attributes $W^{(t)}$ into stable content evidence C , feedback-sensitive hierarchical style $S^{(t)}$, and counterfactual reliability R_{cf} . This reframing turns cross-domain explanation learning into a controlled generation problem: content determines what should be explained, style determines how it is expressed, and reliability determines which counterfactual signals should shape learning.

The resulting method loop combines EASD, HSS, RCR/UCI, and CCV/FCA. It keeps rating evidence content-grounded while allowing the explanation generator to adapt target-domain language without collapsing into a coarse textual shortcut. The current tables provide bounded single-run evidence on strong-related Amazon transfer and weak cross-platform TripAdvisor–Yelp transfer. Future evidence additions can extend this draft with module ablations, multi-seed evaluation, broader baseline adaptation, and human or faithfulness-oriented assessment without changing the central ODCR story.

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